UNIVERSITY OF PADOVA

Embedded Real–Time Control

Laboratory report



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1 Laboratory 1

In this lab ....

1.1 Relevant theoretical notions

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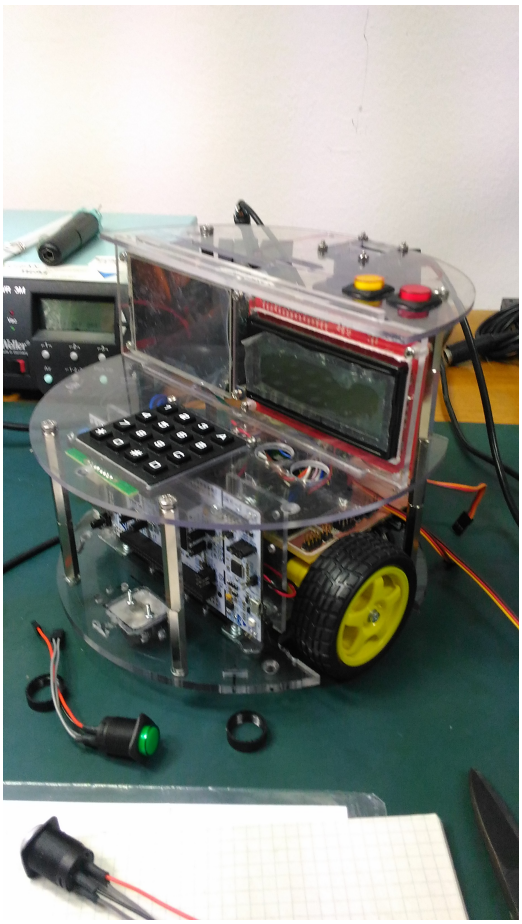


Figure 1: The TurtleBot

In Figure 1 ...

1.2 Brief Introduction and Objectives

Provide a high-level overview of the laboratory experience. Describe the goals of the activity, the hardware or software platform under test, and what key concepts were intended to be explored during the lab session.

1.3 Problem Analysis

Analyse the requirements or specs of the given engineering problem before jumping into implementation:

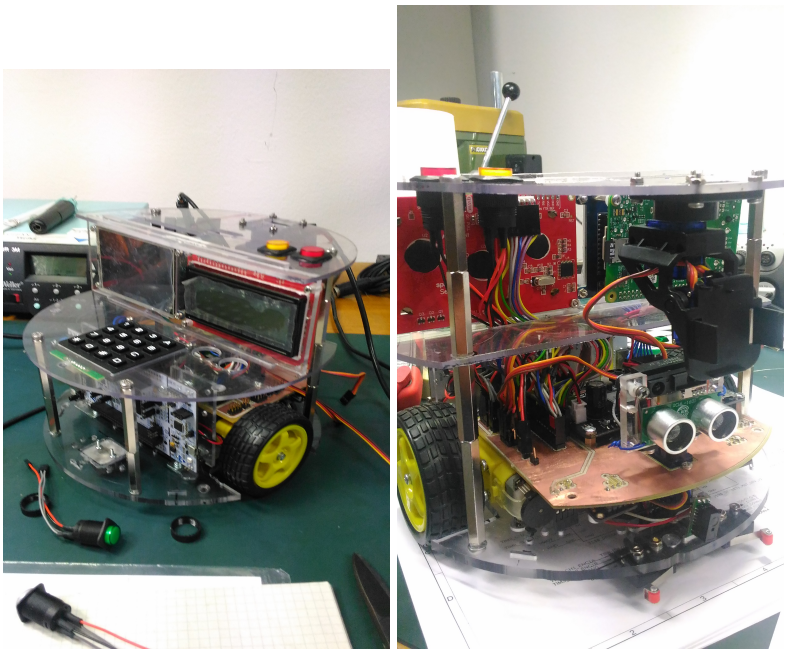
- What are the constraints (e.g., memory limits, sampling rate, processing budget)?
- What performance criteria must be met (e.g., settling time, overshoot, steady-state error)?

1.4 Theoretical Background

Describe the relevant theoretical concepts, mathematical models, equations, or physical laws applied.

For example, considering a continuous transfer function:

$$G(s) = \frac{K}{\tau s + 1} e^{-\theta s} \tag{1}$$



(a) The TurtleBot 1 (b) The TurtleBot 2

Figure 2: The TurtleBot

Explain how this theoretical baseline guides the implementation choice or controller structure.

1.5 Design & Implementation Choices

Detail your design decisions, thought process, and engineering trade-offs.

1.5.1 Tools & Software Environment

- **MATLAB / Simulink:** Used for offline parameter identification and simulation.
- **C / C++ / Embedded IDE:** Used for real-time firmware execution.

1.5.2 Code Structure

Listing 1: C code using listings

```c
#include <stdio.h>
int main()
{
    // print hello to the console
    printf("Hello, world!");
    return 0;
}
```

See code 1.

1.6 Controller Design (If Applicable)

If the activity involved control system synthesis:

- State desired requirements (e.g., Settling Time $T_s < 1.0\,\text{s}$, Phase Margin $\phi_m > 45°$).
- Detail the tuning method (e.g., pole placement, root locus, frequency response tuning).

1.7 Discovered Issues, Calibration & Workarounds

This section captures critical unexpected findings during execution:

- **Unforeseen Issues:** Detail any problems not noticed during the design phase (e.g., noise on sensor pins, numerical overflow).
- **Calibration & Performance Gaps:** If the controller did not achieve perfect calibration or desired performance, explain why (e.g., unmodeled dynamics, saturation limits, transport delay) and propose plausible fixes.
- **Original / Experimental Solutions:** If original or non-standard approaches were tried, describe why they were developed and assess their success or failure.

1.8 Results and Graphical Analysis

Present collected measurements, plots, and experimental data. Describe what each plot demonstrates rather than just attaching graphics.

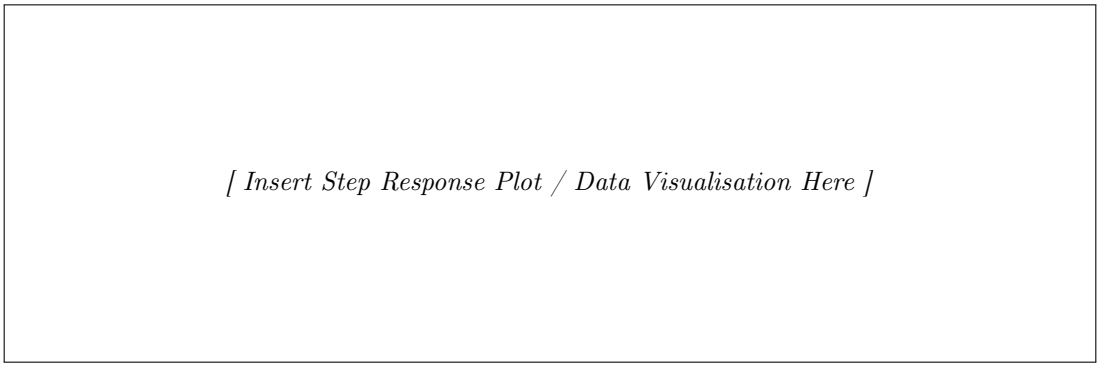


Figure 3: Step response comparison: Target trajectories vs. Measured output.

1.9 Conclusions

Summarise the key learnings, whether the initial objectives were reached, and what improvements could be introduced in future iterations.

A Section in the Appendix

A.1 Subsection in the Appendix

Additional relevant information...